LUMAFLOW · FEASIBILITY ASSESSMENT

LumaFlow feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/lumaflow		
Source	docs/08-feasibility.md @ f88a9c3		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Kit 1.8 upgrade: feasibility of the TRL 3 design (option A), closed on paper

LumaFlow feasibility assessment

Option A is feasible on paper as a garage-built prototype: the science holds (LMF-STM-001, after four detail fixes, two of them found by the independent review), the dose clears 40 mJ/cm² in clear water with an 18 % margin once the 275 nm response factor is applied (below the 20 % the calculation note asks for, so PR-002 is at risk), every part can be made or bought (LMF-BLD-001), and the open questions are measurements for TRL 4, not unknowns that could sink the concept. The weakest points are the liner's wetted reflectance, LED derating (a worst-case stack gives 37.5 mJ/cm²), how well the reactor mixes, the dose rule in firmware, and cost (USD 93 over the value-engineering target). Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Axial reactor with a reflective PTFE-lined bore meets the clear-water dose on paper (RED 50.9 to 74.9 mJ/cm², 47.3 with the 275 nm factor); misses the 70 %/cm test water by design (PR-003, decision D1)	Medium	Dose rests on a liner reflectance of 0.95 (0.80 gives 42.1 with the 275 nm factor), LED derating (S33) and the residence-time distribution (S34), none yet measured	STM S2, S3, S8, S33, S34; CAL B5, B8, B10, B11
Materials	316, PTFE and expanded PTFE, fused silica, EPDM O-rings out of the light, acetal behind PTFE; food-contact grades exist for each	Medium	UV-C through 3 mm of PTFE onto the acetal cap is unmeasured (STM S10); fallback a 316 foil cup	STM S9, S10, S31; REG G2, G4
Manufacturing and assembly	Two machined caps, a TIG-welded boss and a turned holder at a machine shop; the rest is sawing, drilling, tapping and 3D printing; every joint checked in the model (806 contact, clearance and overlap checks pass) and at D3 by `cad_check.py`	High	Ring screws in anti-seize at 0.21 N·m (D-S2); PTFE liner from a specialist supplier	BLD-001; `cad/fidelity.json`; DEC D-S2
Power and controls	22.6 W flowing, 0.23 W standby; an outlet pressure sensor starts the unit when the faucet opens (D-S3), the flow sensor confirms flow; wall photodiode 13.4 nA at 90 %/cm, 11.4 nA at the alarm point, which one failed LED also trips (D-S4)	Medium	The proportional dose rule and the 1 s fail-safe exist only on paper (PR-004 at risk); firmware sketch and a bench test needed at TRL 4	CAL C1 to C5, D2, D5, E1, E2; STM S23 to S26, S37, S38
Supply of parts	Catalogue LEDs, quartz windows, sensors, valves and fittings; one specialist item (high-reflectance PTFE tube)	Medium	LED output bin and price at quantity; flow coefficients of the sensor and valve; liner reflectance data from the supplier	BOM; DEC To confirm 2, 4, 7
Cost	USD 433 against a USD 340 value-engineering target	Medium	Drivers: LEDs, 316 cap, liner, tube, window; savings listed in DEC	CAL I1; DEC Value engineering

Ratings: High, Medium, Low.